

Problem Set 4 – Function families

Session 4 · Polynomials, exponentials, logarithms, and trigonometric models

Part A · Mechanics

1 POLYNOMIAL SHAPE

Let $p(x) = -x^2 + 2x + 3$.

- State the degree and leading coefficient of p .
- Describe what the graph does at its left and right ends.
- Sketch the graph. Give the intercepts with both axes and the vertex, and say how you found each one.

2 QUADRATIC TRANSFORMATION

Start with $f(x) = x^2$. The function

$$g(x) = -(x - 2)^2 + 4$$

is made from f .

- Describe the shift and reflection.
- Give the vertex and find the real roots of g .

3 EXPONENTIAL EQUATION

Solve

$$5^{2x} - 30 \cdot 5^x + 125 = 0.$$

Give all real solutions and justify each step.

4 CHANGE OF BASE

Solve exactly for x :

$$\log_2(x - 1) + \log_4(x - 1) = 6.$$

Use the change-of-base formula, and state the domain restriction before solving.

Part B · Reasoning and modelling

5 LOGARITHMIC EQUATION

Solve over the real numbers:

$$\log_2(x - 1) + \log_2(x + 1) = 3.$$

State the domain before combining the logarithms. Check every possible answer in the original equation.

6 GROWTH COMPARISON

Put the following functions in order from slowest growth to fastest growth as the positive integer n becomes very large:

$$\log_2 n, \quad n, \quad n \log_2 n, \quad n^2, \quad 2^n.$$

In two or three sentences, explain why checking one value such as $n = 3$ does not prove the full ranking.

7 PERIODIC MODEL

Let t be the number of hours after midnight. A simple model for temperature is

$$T(t) = 3 \sin\left(\frac{\pi}{6}(t - 1)\right) + 20 \quad (0 \leq t \leq 12).$$

- State the amplitude, midline, period, and range.
- At what time does the model first reach a maximum? At what time does it reach a minimum?
- Explain why $(t - 1)$ moves the graph right by 1 hour, rather than left by 1 hour.

Part C · Communicate precisely

8 FIND THE ERROR

A student writes: " $\log_2(u + v) = \log_2 u + \log_2 v$ because logarithms turn an operation into addition."

- Identify the false statement.
- Give a counterexample with positive u and v .
- Write the correct logarithm law for a product, including its condition.

The statement “An exponential function is always bigger than a polynomial” is false or unclear as written. Rewrite it as a true statement about c^n and n^k , including all important conditions. Then give one numerical example showing why the words “for sufficiently large n ” are necessary.
